

Lab 18: Mini Project: Investigating Pertussis Resurgence

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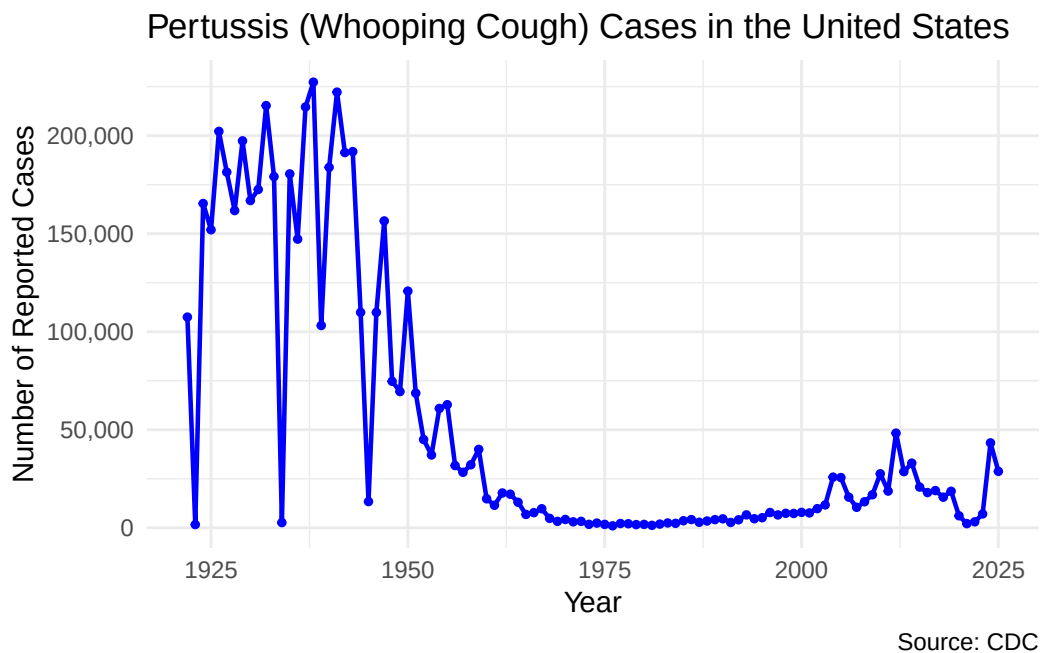
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```
library(datapasta)
library(ggplot2)
```

Q1. With the help of the R “addin” package datapasta assign the CDC pertussis case number data to a data frame called cdc and use ggplot to make a plot of cases numbers over time.

```
cdc <- data.frame(
  Year = c(2025,2024,2023,2022,2021,2020,2019,2018,2017,2016,2015,2014,2013,2012,2011,
    2010,2009,2008,2007,2006,2005,2004,2003,2002,2001,2000,1999,1998,1997,1996,
    1995,1994,1993,1992,1991,1990,1989,1988,1987,1986,1985,1984,1983,1982,1981,
    1980,1979,1978,1977,1976,1975,1974,1973,1972,1971,1970,1969,1968,1967,1966,
    1965,1964,1963,1962,1961,1960,1959,1958,1957,1956,1955,1954,1953,1952,1951,
    1950,1949,1948,1947,1946,1945,1944,1943,1942,1941,1940,1939,1938,1937,1936,
    1935,1934,1933,1932,1931,1930,1929,1928,1927,1926,1925,1924,1923,1922),
  Cases = c(28783,43321,7063,3044,2116,6124,18617,15609,18975,17972,20762,32971,28639,
    48277,18719,27550,16858,13278,10454,15632,25616,25827,11647,9771,7580,7867,
    7298,7405,6564,7796,5137,4617,6586,4083,2719,4570,4157,3450,2823,4195,3589,
    2276,2463,1895,1248,1730,1623,2063,2177,1010,1738,2402,1759,3287,3036,4249,
    3285,4810,9718,7717,6799,13005,17135,17749,11468,14809,40005,32148,28295,
    31732,62786,60886,37129,45030,68687,120718,69479,74715,156517,109860,13379,
    109873,191890,191383,222202,183866,103188,227319,214652,147237,180518,2652,
    179135,215343,172559,166914,197371,161799,181411,202210,152003,165418,1641,107473)
)
```

```
ggplot(cdc, aes(x = Year, y = Cases)) +
  geom_line(color = "blue", linewidth = 0.8) +
  geom_point(color = "blue", size = 1) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Pertussis (Whooping Cough) Cases in the United States",
    x = "Year",
    y = "Number of Reported Cases",
    caption = "Source: CDC"
  ) +
  theme_minimal()
```



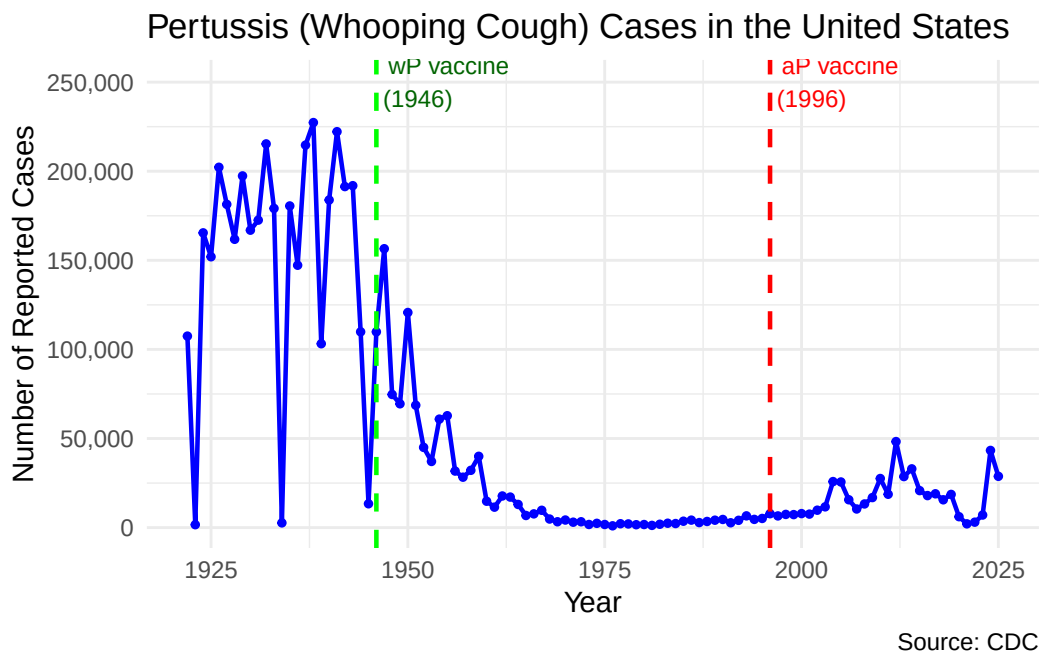
Q2. Using the ggplot `geom_vline()` function add lines to your previous plot for the 1946 introduction of the wP vaccine and the 1996 switch to aP vaccine (see example in the hint below). What do you notice?

```
ggplot(cdc, aes(x = Year, y = Cases)) +
  geom_line(color = "blue", linewidth = 0.8) +
  geom_point(color = "blue", size = 1) +
  geom_vline(xintercept = 1946, linetype = "dashed", color = "green", linewidth = 0.8) +
  geom_vline(xintercept = 1996, linetype = "dashed", color = "red", linewidth = 0.8) +
  annotate("text", x = 1946, y = 250000, label = "wP vaccine\n(1946)",
```

```

    hjust = -0.1, color = "darkgreen", size = 3) +
  annotate("text", x = 1996, y = 250000, label = "aP vaccine\n(1996)",
    hjust = -0.1, color = "red", size = 3) +
  scale_y_continuous(labels = scales::comma) +
  labs(
    title = "Pertussis (Whooping Cough) Cases in the United States",
    x = "Year",
    y = "Number of Reported Cases",
    caption = "Source: CDC"
  ) +
  theme_minimal()

```



Q3. Describe what happened after the introduction of the aP vaccine? Do you have a possible explanation for the observed trend?

After the introduction of the aP vaccine in 1996, pertussis cases gradually rebounded with major spikes around 2010 and 2012. This is likely because the aP vaccine provides shorter lasting immunity than the older wP vaccine, allowing cases to rise as protection fades over time.

```
library(jsonlite)
```

```
subject <- read_json("https://www.cmi-pb.org/api/subject", simplifyVector = TRUE)
```

```
head(subject, 3)
```

```

  subject_id infancy_vac biological_sex ethnicity race
1          1          wP      Female Not Hispanic or Latino White
2          2          wP      Female Not Hispanic or Latino White
3          3          wP      Female      Unknown White
  year_of_birth date_of_boost   dataset
1  1986-01-01   2016-09-12 2020_dataset
2  1968-01-01   2019-01-28 2020_dataset
3  1983-01-01   2016-10-10 2020_dataset

```

Q4. How many aP and wP infancy vaccinated subjects are in the dataset?

```
table(subject$infancy_vac)
```

```

aP wP
87 85

```

Q5. How many Male and Female subjects/patients are in the dataset?

```
table(subject$biological_sex)
```

```

Female   Male
  112     60

```

Q6. What is the breakdown of race and biological sex (e.g. number of Asian females, White males etc...)?

```
table(subject$race, subject$biological_sex)
```

	Female	Male
American Indian/Alaska Native	0	1
Asian	32	12
Black or African American	2	3
More Than One Race	15	4
Native Hawaiian or Other Pacific Islander	1	1
Unknown or Not Reported	14	7
White	48	32

```
library(lubridate)
```

Attaching package: 'lubridate'

The following objects are masked from 'package:base':

date, intersect, setdiff, union

```
today()
```

```
[1] "2026-06-01"
```

```
today() - ymd("2000-01-01")
```

Time difference of 9648 days

```
time_length( today() - ymd("2000-01-01"), "years")
```

```
[1] 26.41478
```

Q7. Using this approach determine (i) the average age of wP individuals, (ii) the average age of aP individuals; and (iii) are they significantly different?

```
subject$age <- today() - ymd(subject$year_of_birth)
```

```
tapply(time_length(subject$age, "years"), subject$infancy_vac, mean)
```

aP	wP
28.31013	37.06084

```
t.test(time_length(age, "years") ~ infancy_vac, data = subject)
```

Welch Two Sample t-test

```
data: time_length(age, "years") by infancy_vac
t = -12.918, df = 104.03, p-value < 2.2e-16
alternative hypothesis: true difference in means between group aP and group wP is not equal to 0
95 percent confidence interval:
 -10.094058 -7.407351
sample estimates:
mean in group aP mean in group wP
      28.31013      37.06084
```

Q8. Determine the age of all individuals at time of boost?

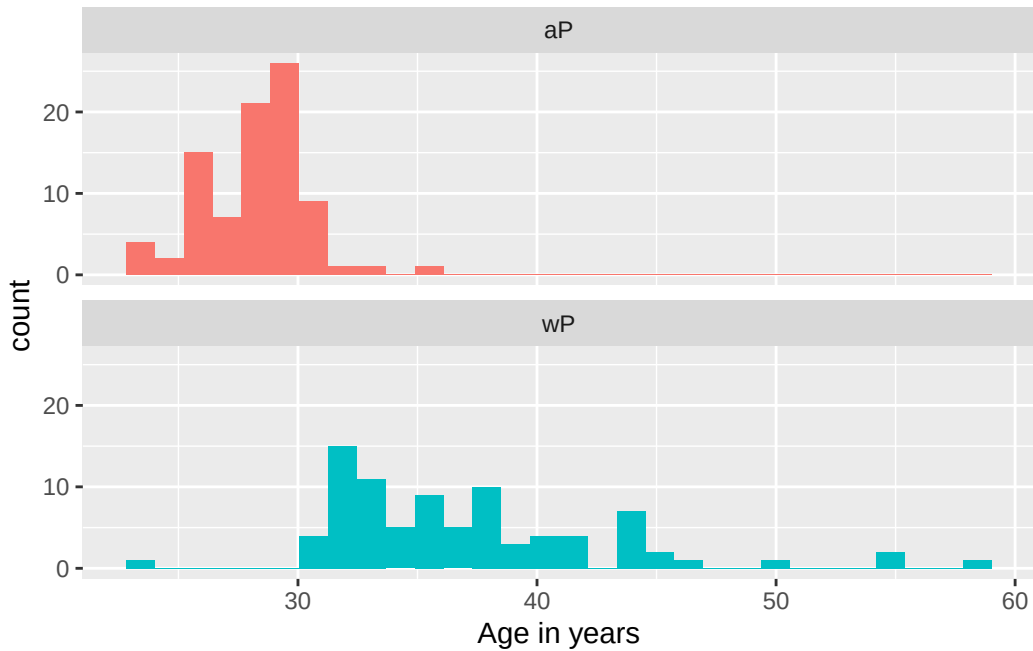
```
subject$age_at_boost <- time_length(ymd(subject$date_of_boost) - ymd(subject$year_of_birth),
head(subject$age_at_boost)
```

```
[1] 30.69678 51.07461 33.77413 28.65982 25.65914 28.77481
```

Q9. With the help of a faceted boxplot or histogram (see below), do you think these two groups are significantly different?

```
ggplot(subject) +
  aes(time_length(age, "year"), fill = as.factor(infancy_vac)) +
  geom_histogram(show.legend = FALSE) +
  facet_wrap(vars(infancy_vac), nrow = 2) +
  xlab("Age in years")
```

`stat\_bin()` using `bins = 30`. Pick better value `binwidth`.



```
library(dplyr)
```

Attaching package: 'dplyr'

The following objects are masked from 'package:stats':

filter, lag

The following objects are masked from 'package:base':

intersect, setdiff, setequal, union

```
library(jsonlite)
```

```
specimen <- read_json("https://www.cmi-pb.org/api/specimen", simplifyVector = TRUE)
titer <- read_json("https://www.cmi-pb.org/api/plasma_ab_titer", simplifyVector = TRUE)
```

Q9. Complete the code to join specimen and subject tables to make a new merged data frame containing all specimen records along with their associated subject details:

```
meta <- inner_join(specimen, subject, by = "subject_id")
dim(meta)
```

```
[1] 1503    15
```

```
head(meta)
```

```
specimen_id subject_id actual_day_relative_to_boost
1           1           1                        -3
2           2           1                         1
3           3           1                         3
4           4           1                         7
5           5           1                        11
6           6           1                        32
planned_day_relative_to_boost specimen_type visit infancy_vac biological_sex
1                             0         Blood      1          wP         Female
2                             1         Blood      2          wP         Female
3                             3         Blood      3          wP         Female
4                             7         Blood      4          wP         Female
5                            14         Blood      5          wP         Female
6                            30         Blood      6          wP         Female
ethnicity race year_of_birth date_of_boost dataset
1 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
2 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
3 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
4 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
5 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
6 Not Hispanic or Latino White 1986-01-01 2016-09-12 2020_dataset
age age_at_boost
1 14761 days 30.69678
2 14761 days 30.69678
3 14761 days 30.69678
4 14761 days 30.69678
5 14761 days 30.69678
6 14761 days 30.69678
```

Q10. Now using the same procedure join meta with titer data so we can further analyze this data in terms of time of visit aP/wP, male/female etc.

```
abdata <- inner_join(titer, meta, by = "specimen_id")
dim(abdata)
```



```
[1] 52576    22
```

```
names(abdata)
```

```
[1] "specimen_id"          "isotype"
[3] "is_antigen_specific"  "antigen"
[5] "MFI"                  "MFI_normalised"
[7] "unit"                 "lower_limit_of_detection"
[9] "subject_id"           "actual_day_relative_to_boost"
[11] "planned_day_relative_to_boost" "specimen_type"
[13] "visit"                "infancy_vac"
[15] "biological_sex"       "ethnicity"
[17] "race"                 "year_of_birth"
[19] "date_of_boost"        "dataset"
[21] "age"                  "age_at_boost"
```

Q11. How many specimens (i.e. entries in abdata) do we have for each isotype?

```
table(abdata$isotype)
```

```
  IgE   IgG  IgG1  IgG2  IgG3  IgG4
6698  5389 10117 10124 10124 10124
```

Q12. What are the different \$dataset values in abdata and what do you notice about the number of rows for the most “recent” dataset?

```
table(abdata$dataset)
```

```
2020_dataset 2021_dataset 2022_dataset 2023_dataset
          31520           8085           7301           5670
```

```
igg <- abdata %>% filter(isotype == "IgG")
head(igg)
```

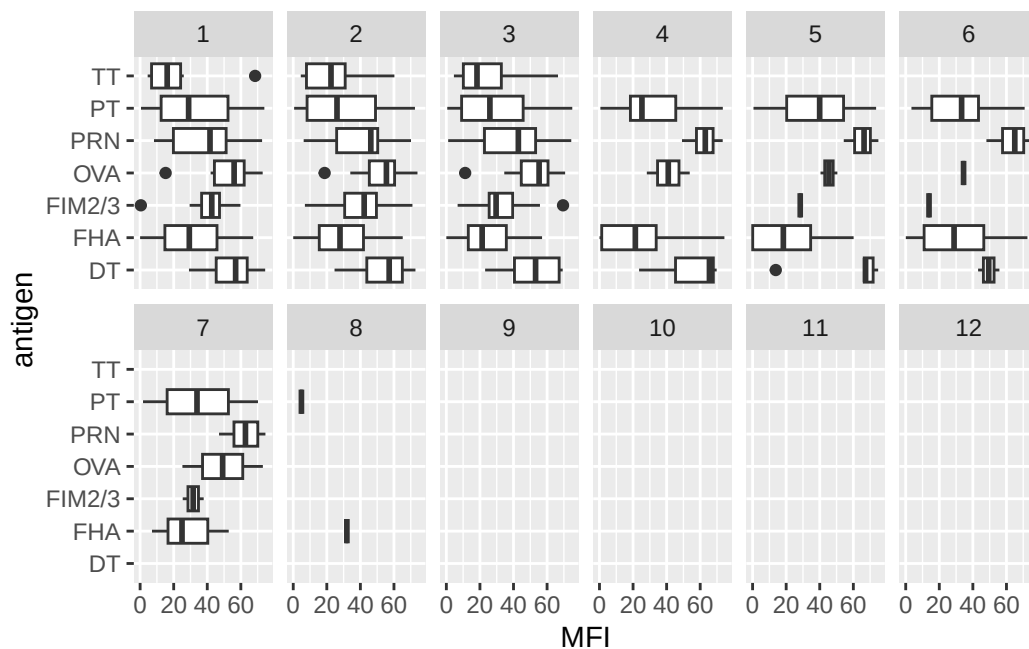
```
specimen_id isotype is_antigen_specific antigen      MFI MFI_normalised
1           1     IgG                TRUE      PT  68.56614      3.736992
2           1     IgG                TRUE      PRN 332.12718      2.602350
3           1     IgG                TRUE      FHA 1887.12263     34.050956
```

4	19	IgG	TRUE	PT	20.11607	1.096366
5	19	IgG	TRUE	PRN	976.67419	7.652635
6	19	IgG	TRUE	FHA	60.76626	1.096457
	unit	lower_limit_of_detection	subject_id	actual_day_relative_to_boost		
1	IU/ML	0.530000	1	-3		
2	IU/ML	6.205949	1	-3		
3	IU/ML	4.679535	1	-3		
4	IU/ML	0.530000	3	-3		
5	IU/ML	6.205949	3	-3		
6	IU/ML	4.679535	3	-3		
	planned_day_relative_to_boost	specimen_type	visit	infancy_vac	biological_sex	
1	0	Blood	1	wP	Female	
2	0	Blood	1	wP	Female	
3	0	Blood	1	wP	Female	
4	0	Blood	1	wP	Female	
5	0	Blood	1	wP	Female	
6	0	Blood	1	wP	Female	
	ethnicity	race	year_of_birth	date_of_boost	dataset	
1	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
2	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
3	Not Hispanic or Latino	White	1986-01-01	2016-09-12	2020_dataset	
4	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
5	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
6	Unknown	White	1983-01-01	2016-10-10	2020_dataset	
	age	age_at_boost				
1	14761 days	30.69678				
2	14761 days	30.69678				
3	14761 days	30.69678				
4	15857 days	33.77413				
5	15857 days	33.77413				
6	15857 days	33.77413				

Q13. Complete the following code to make a summary boxplot of Ab titer levels (MFI) for all antigens:

```
ggplot(igg) +
  aes(MFI, antigen) +
  geom_boxplot() +
  xlim(0, 75) +
  facet_wrap(vars(visit), nrow = 2)
```

Warning: Removed 4475 rows containing non-finite outside the scale range (`stat\_boxplot()`).

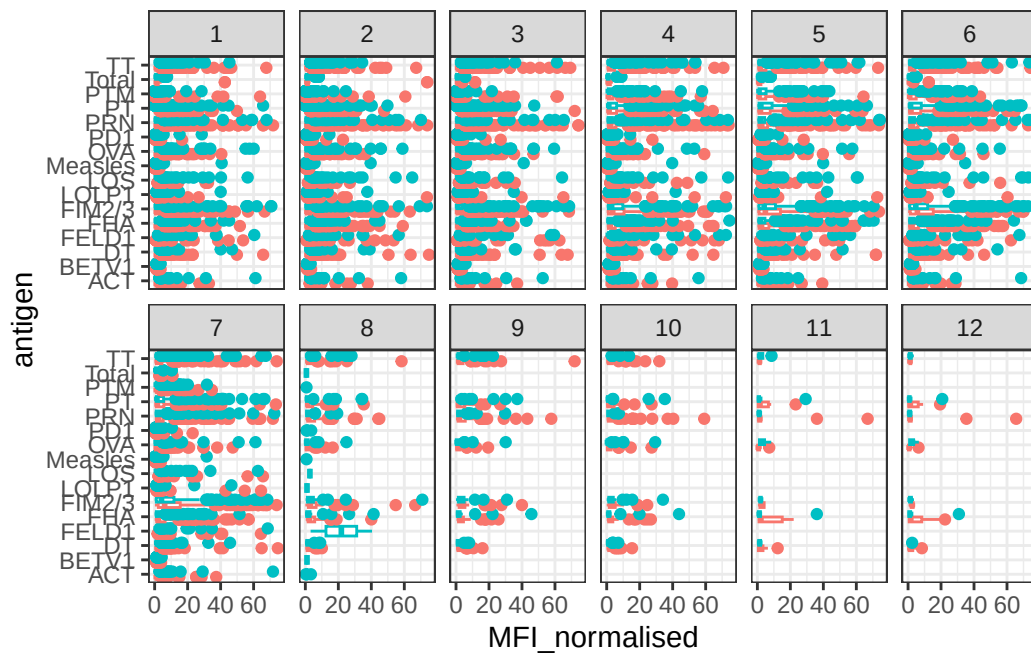


Q14. What antigens show differences in the level of IgG antibody titers recognizing them over time? Why these and not others?

FIM2/3, FHA, PRN, and PT show the clearest increases in IgG titer levels over time, particularly after the booster vaccination. These antigens are directly targeted by the pertussis vaccine, unlike unrelated antigens such as TT and DT. Also controls like OVA.

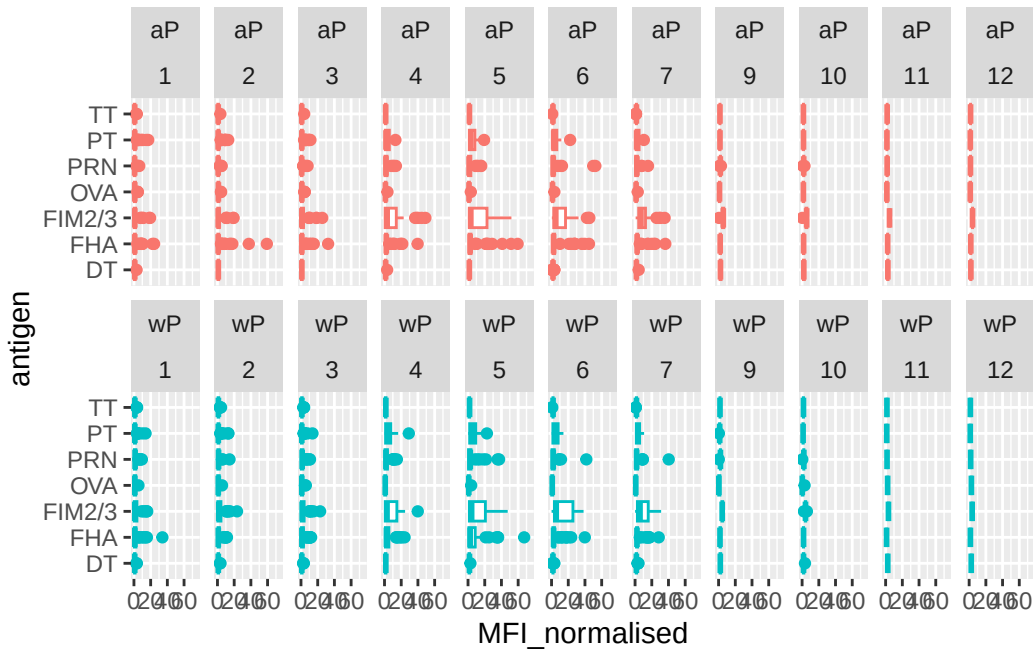
```
ggplot(abdata) +
  aes(x = MFI_normalised, y = antigen, col = infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit), nrow = 2) +
  xlim(0, 75) +
  theme_bw()
```

Warning: Removed 915 rows containing non-finite outside the scale range (``stat_boxplot()``).



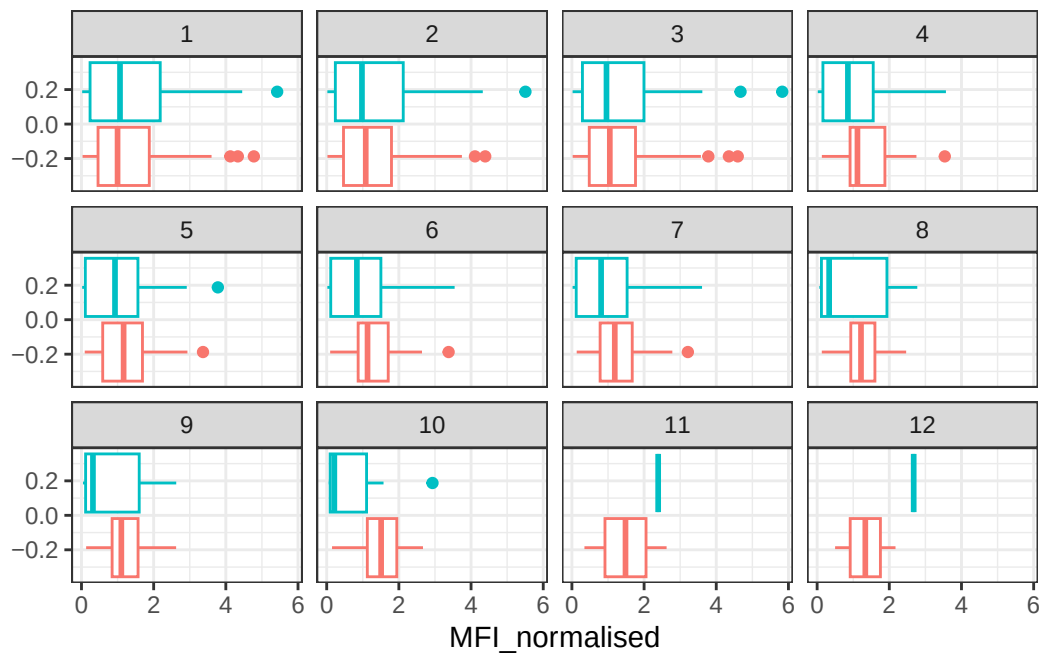
```
igg %>% filter(visit != 8) %>%
ggplot() +
  aes(MFI_normalised, antigen, col=infancy_vac ) +
  geom_boxplot(show.legend = FALSE) +
  xlim(0,75) +
  facet_wrap(vars(infancy_vac, visit), nrow=2)
```

Warning: Removed 5 rows containing non-finite outside the scale range (`stat\_boxplot()`).

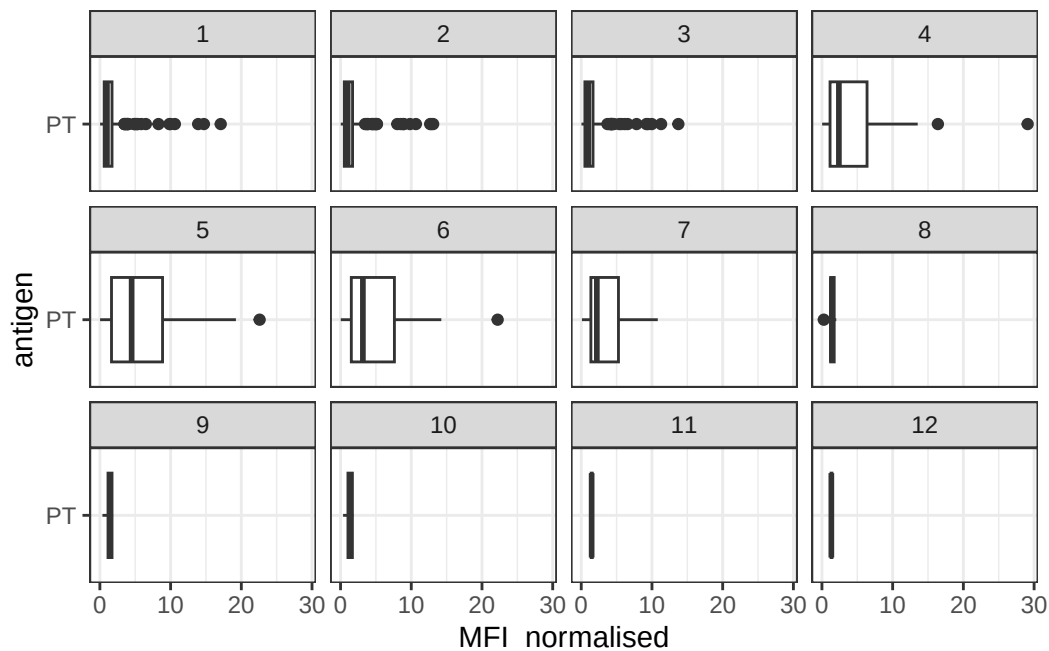


Q15. Filter to pull out only two specific antigens for analysis and create a boxplot for each. You can chose any you like. Below I picked a “control” antigen (“OVA”, that is not in our vaccines) and a clear antigen of interest (“PT”, Pertussis Toxin, one of the key virulence factors produced by the bacterium *B. pertussis*).

```
filter(igg, antigen=="OVA") %>%
  ggplot() +
  aes(MFI_normalised, col=infancy_vac) +
  geom_boxplot(show.legend = FALSE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



```
filter(igg, antigen == "PT") %>%
  ggplot() +
  aes(MFI_normalised, antigen) +
  geom_boxplot(show.legend = TRUE) +
  facet_wrap(vars(visit)) +
  theme_bw()
```



Q16. What do you notice about these two antigens time courses and the PT data in particular?

OVA shows no change across visits as expected, while PT rises dramatically after boosting with wP individuals showing higher IgG responses than aP individuals.

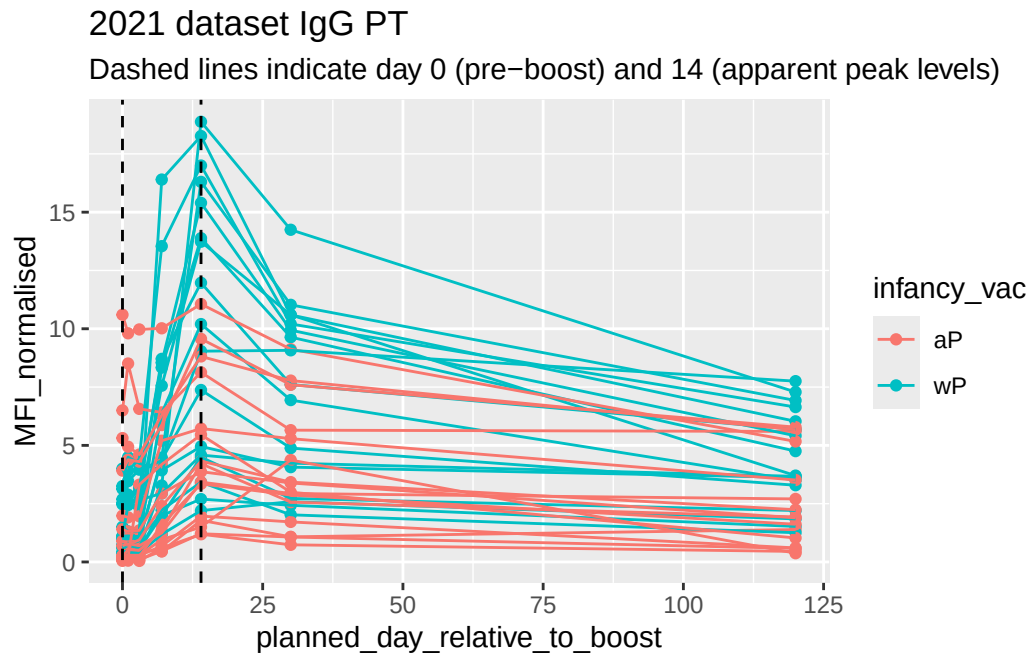
Q17. Do you see any clear difference in aP vs. wP responses?

Yes, wP primed individuals consistently show higher PT antibody titers after boosting, suggesting the whole cell vaccine primes a stronger long term immune memory than the acellular vaccine.

```
abdata.21 <- abdata %>% filter(dataset == "2021_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
         y=MFI_normalised,
         col=infancy_vac,
         group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
```

```
geom_vline(xintercept=14, linetype="dashed") +
labs(title="2021 dataset IgG PT",
      subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```

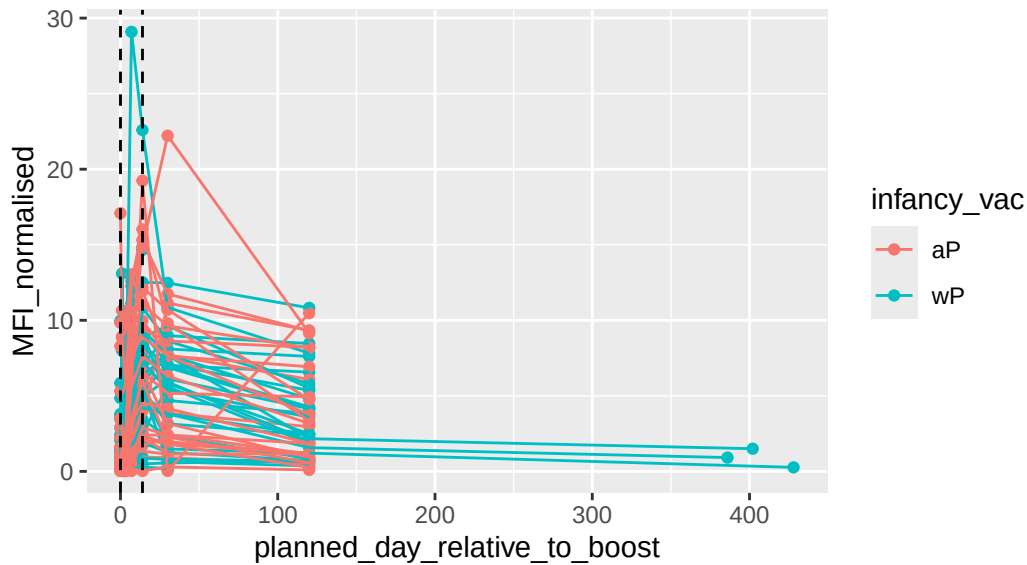


```
abdata.21 <- abdata %>% filter(dataset == "2020_dataset")

abdata.21 %>%
  filter(isotype == "IgG", antigen == "PT") %>%
  ggplot() +
    aes(x=planned_day_relative_to_boost,
         y=MFI_normalised,
         col=infancy_vac,
         group=subject_id) +
    geom_point() +
    geom_line() +
    geom_vline(xintercept=0, linetype="dashed") +
    geom_vline(xintercept=14, linetype="dashed") +
    labs(title="2021 dataset IgG PT",
          subtitle = "Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)")
```


2021 dataset IgG PT

Dashed lines indicate day 0 (pre-boost) and 14 (apparent peak levels)



Q18. Does this trend look similar for the 2020 dataset?

Yes. The 2020 dataset shows a similar trend: PT IgG levels peak around day 14 post boost then decline.

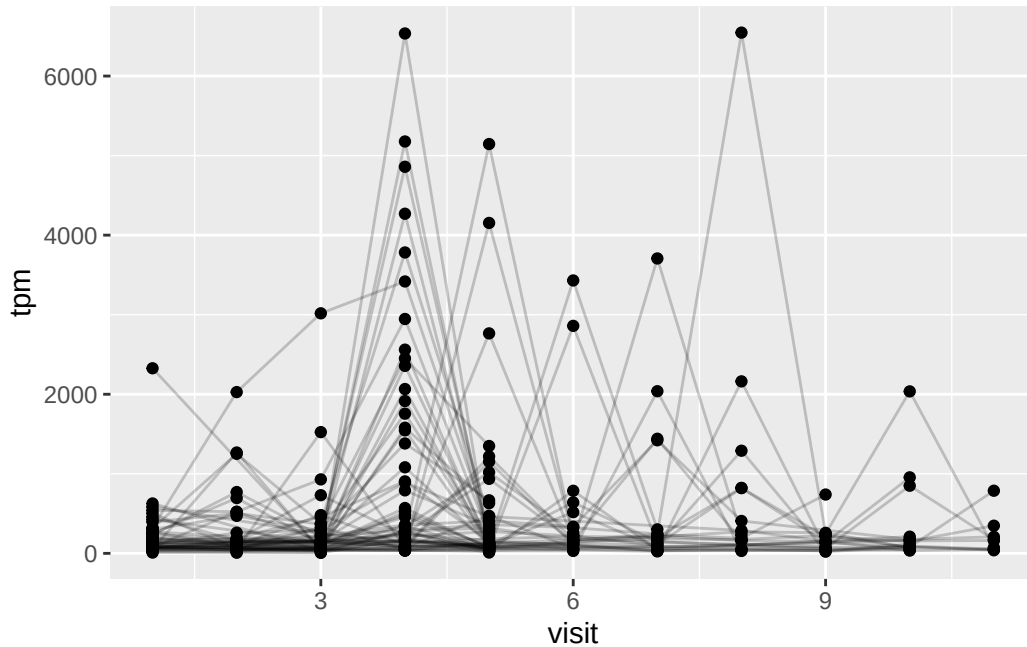
```
url <- "https://www.cmi-pb.org/api/v2/rnaseq?versioned_ensembl_gene_id=eq.ENSNG00000211896.7"
rna <- read_json(url, simplifyVector = TRUE)
```

```
ssrna <- inner_join(rna, meta)
```

Joining with `by = join\_by(specimen\_id)`

Q19. Make a plot of the time course of gene expression for IGHG1 gene (i.e. a plot of visit vs. tpm).

```
ggplot(ssrna) +
  aes(visit, tpm, group=subject_id) +
  geom_point() +
  geom_line(alpha=0.2)
```



Q20.: What do you notice about the expression of this gene (i.e. when is it at it's maximum level)?

IGHG1 expression peaks dramatically at visit 4 before dropping back down to baseline levels.

Q21. Does this pattern in time match the trend of antibody titer data? If not, why not?

Yes. This matches the antibody titer trend, which makes sense since IGHG1 encodes an IgG antibody protein. Increased gene expression precedes and drives the rise in detectable IgG antibody levels.